EXP 11: Write a program for the task of Credit Score Classification

CODING:

```
# Credit Score Classification using KNN

import math

# Training Dataset
X = [
    [1, 25000],
    [2, 30000],
    [3, 45000],
    [4, 60000],
    [5, 80000],
    [6, 100000],
    [2, 35000],
    [5, 75000]
]

Y = [
    'Low', 'Low', 'Medium', 'Medium',
    'High', 'High', 'Low', 'High'
]

# Euclidean Distance
def distance(a, b):
    total = 0
```

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```
    for i in range(len(a)):
        total += (a[i] - b[i]) ** 2

    return math.sqrt(total)

# KNN Classification
def predict(sample, k=3):
    distances = []

    for i in range(len(X)):
        d = distance(sample, X[i])
        distances.append((d, Y[i]))

    distances.sort(key=lambda x: x[0])

    neighbours = []

    for i in range(k):
        neighbours.append(distances[i][1])

    # Find Majority Class
    prediction = max(set(neighbours),
                     key=neighbours.count)

    return prediction

# Display Training Data
print("Credit Score Classification")
```

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```
print("\nTraining Dataset:")


for i in range(len(X)):
    print(X[i], ":", Y[i])

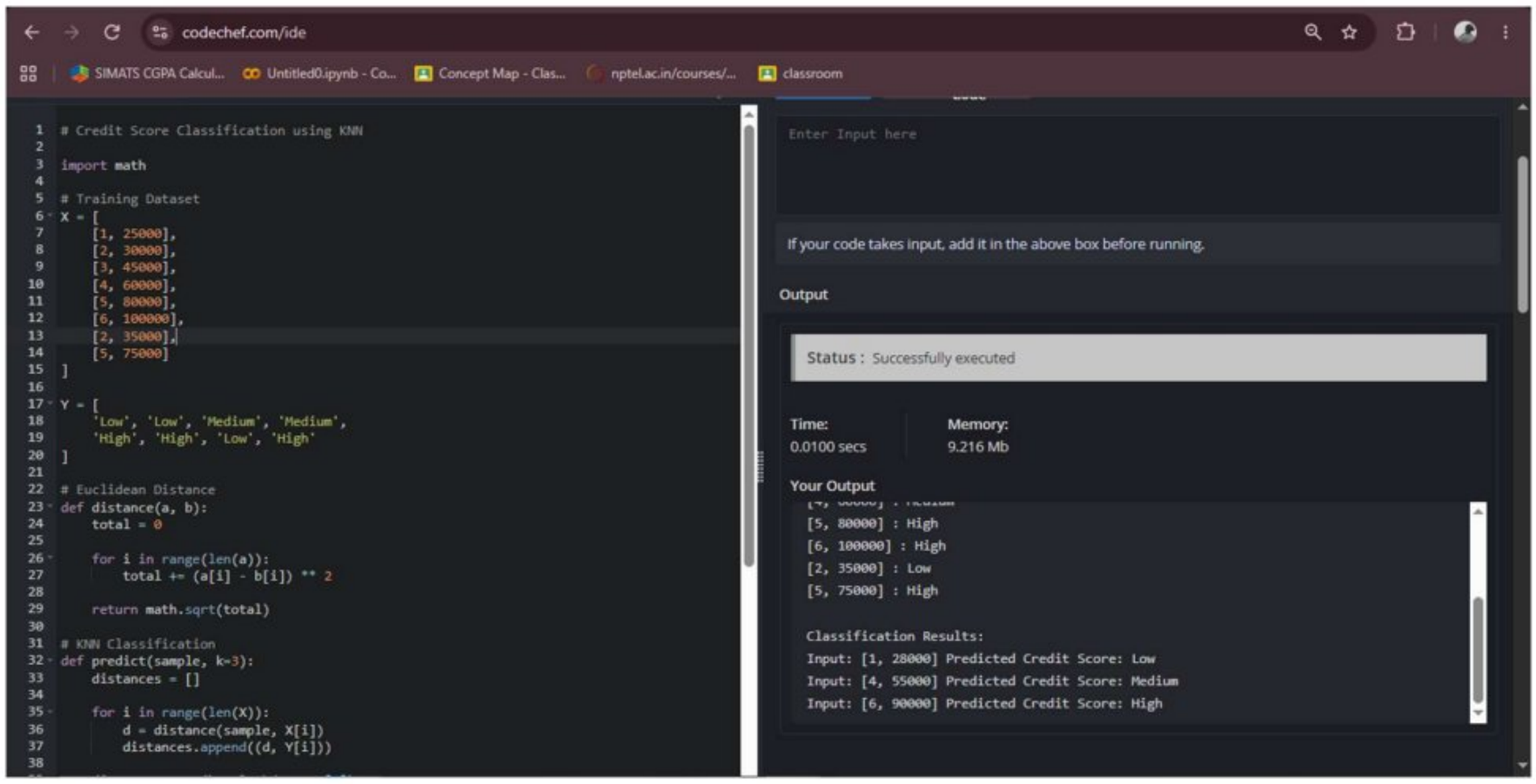

# Test Data
test_data = [
    [1, 28000],
    [4, 55000],
    [6, 90000]
]


print("\nClassification Results:")


for sample in test_data:
    result = predict(sample)
    print("Input:", sample,
          "Predicted Credit Score:", result)
```

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OUTPUT:




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